3.7 PD-WEIGHTED & T1-WEIGHTED SEQUENCES

ANATOMY, PATHOLOGY, AND CONTRAST ENHANCEMENT

PD-weighted images show anatomy with low contrast and low sensitivity to T1 effects.
T1-weighted images highlight differences in T1 relaxation—excellent for anatomy and after gadolinium.

KEY CONCEPT

- PD = Proton Density (reflects number of hydrogen protons).
- T1 weighting = Short TR and TE → emphasizes T1 differences.
- PD weighting = Long TR and short (or moderate) TE → minimizes T1 effects.
- Use PD for anatomy and subtle pathology; use T1 for anatomy, fat, and post-contrast imaging.

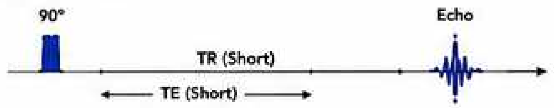
↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW THEY WORK

A. PD-WEIGHTED

- Long TR minimizes T1 differences.
- Short or moderate TE minimizes T2 weighting.
- Signal reflects proton density (PD) → tissue water content.

B. T1-WEIGHTED



- Short TR preserves T1 differences.
- Short TE minimizes T2 effects.
- Signal reflects T1 relaxation → fat bright, fluid dark.

★ PD = anatomy with low contrast | T1 = anatomy with high contrast

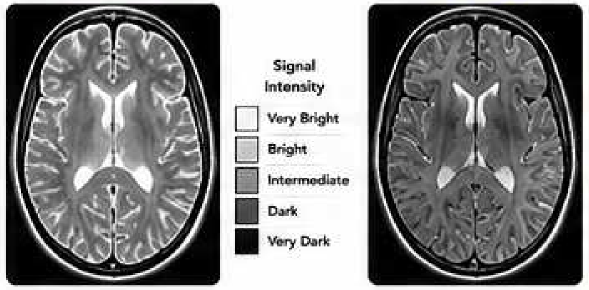
2 KEY DIFFERENCES

Feature	PD-Weighted	T1-Weighted
TR	Long (≥ 2000 ms)	Short (300–800 ms)
TE	Short / Moderate (10–30 ms)	Short (10–15 ms)
T1 Weighting	Minimal	High
T2 Weighting	Low	Low
PD Weighting	High	Low
Fat Signal	Intermediate	Bright
Water / CSF	Bright	Dark
Contrast	Low	High
Scan Time	Longer	Shorter
Best Use	Anatomy, joint space, subtle pathology, quantitative work	Anatomy, fat, post-contrast, lesion detection

ⓘ PD emphasizes "how much" (protons).
T1 emphasizes "how fast" (recovery).

3 IMAGE APPEARANCE COMPARISON (1.5T BRAIN)

PD-Weighted | T1-Weighted



PD-Weighted:
- CSF: Bright
- White Matter: Bright
- Gray Matter: Intermediate
- Fat (scalp): Intermediate
- Lesions: Variable (often bright)

T1-Weighted:
- CSF: Dark
- White Matter: Bright
- Gray Matter: Intermediate
- Fat (scalp): Bright
- Lesions: Variable (often dark)

ⓘ Appearance depends on TR, TE, flip angle, field strength, and coil.

4 WHEN TO USE WHICH?

Clinical Goal / Scenario	Prefer	Why?
Routine anatomy (brain, spine)	T1	Better anatomic detail, shorter scan, fat bright, CSF dark.
Joint anatomy (meniscus, labrum, ligaments, cartilage)	PD (FS)	High tissue contrast, fluid bright, evaluates cartilage and edema.
Subtle pathology (edema, marrow change, demyelination)	PD or PD-FS	High PD sensitivity; shows fluid and water content.
Post-contrast imaging	T1	Gadolinium shortens T1 → lesions enhance (bright).
Fat evaluation	T1	Fat is naturally bright—helps identify and differentiate.
Quantitative / research	PD	More reproducible, less biased by T1 differences.
Short scan time needed	T1	Short TR decreases scan time.

5 EFFECT OF PARAMETER CHANGES

PD-Weighted (Increase PD Weighting)

Parameter	Change	Effect
TR	Increase	More PD weighting, less T1
TE	Shorten	Less T2 weighting
Flip Angle	Lower	Less T1 weighting
Bandwidth	Increase	Less chemical shift, less artifacts

T1-Weighted (Increase T1 Weighting)

Parameter	Change	Effect
TR	Decrease	More T1 weighting
TE	Shorten	Less T2 weighting
Flip Angle	Increase	More T1 weighting, higher SNR
Bandwidth	Decrease	Higher SNR (but more artifacts)

⚙ Balance TR, TE, and flip angle for optimal contrast and SNR.

6 TISSUE SIGNAL SUMMARY (1.5T)

Tissue	PD-Weighted	T1-Weighted
Fat	Intermediate	Very Bright
CSF / Water	Very Bright	Very Dark
White Matter	Bright	Bright
Gray Matter	Intermediate	Intermediate
Muscle	Intermediate	Intermediate
Cortical Bone	Very Dark	Very Dark
Air	Very Dark	Very Dark
Enhancing Lesion (Post-Gadolinium)	Variable	Bright

ⓘ Gadolinium shortens T1 → tissues with gad uptake appear bright on T1.

7 PD-FS vs T1-FS (WITH FAT SATURATION)

PD-FS

- Fat suppressed
- Fluid / edema: Bright
- Cartilage: Bright
- Excellent for edema and joint pathology

T1-FS (Post-Contrast)

- Fat suppressed
- Fluid / edema: Darker
- Enhancement: Bright
- Best for detecting enhancing lesions

ⓘ Fat sat (spectral) reduces chemical shift and improves lesion conspicuity.

8 ENGINEER CHECKLIST (PD & T1)

- ☑ Is TR / TE appropriate for the desired contrast?
- ☑ Is fat bright on T1 and not overly bright on PD?
- ☑ Is CSF dark on T1 and bright on PD?
- ☑ Is chemical shift artifact minimal? Check BW / CF.
- ☑ Is coil selection optimal for SNR and coverage?
- ☑ Is uniformity good across FOV?
- ☑ Are shimming and center frequency correct?
- ☑ Any motion degrading contrast or sharpness?
- ☑ Repeat test or sequence if image quality suboptimal.

QUICK REFERENCE PD vs T1

TR	Long	Short
TE	Short/Mod	Short
Weighting	PD	T1
CSF	Bright	Dark
Fat	Intermed.	Bright
Best For	Anatomy Subtle Path	Anatomy Post-Contrast

9 CLINICAL EXAMPLES

	PD	T1
Brain	White matter disease, edema, demyelination, MS plaques	Anatomy, pituitary, post-contrast lesions, tumors
Spine	Disc hydration, endplate edema, ligament / tendon evaluation	Anatomy, post-contrast, marrow lesions, metastases
Knee	Meniscus, cartilage, ACL/PCL, marrow edema, effusion	Anatomy, fat planes, post-contrast synovitis, tumors
Body	Fluid collections, cysts, edema, organ contrast	Anatomy, fat, post-contrast lesions, perfusion studies

10 PARAMETERS THAT MATTER MOST

Parameter	Impact on PD	Impact on T1
TR	Most Important (Increase)	Most Important (Decrease)
TE	High (Shorten)	Moderate (Shorten)
Flip Angle	Moderate (Lower)	High (Higher)
Bandwidth	Lower BW = More SNR (higher artifacts)	Lower BW = More SNR (higher artifacts)
Field Strength	Higher field = More SNR & more T1 differences	Higher field = More SNR & more T1 differences

11 PRACTICAL TIPS

- Use PD-FS for edema and subtle pathology.
- Use T1 before and after contrast for lesion detection.
- Shorter TR = stronger T1; longer TR = more PD.
- Always check fat signal to confirm correct weighting.
- Optimize BW and fat sat for clean, artifact-free images.
- Compare with other planes and sequences.

REMEMBER

PD-weighted = anatomy with low contrast.
T1-weighted = anatomy with high contrast.
T1 + Gadolinium = highlights disease.
Choose the right sequence for the right question.

KEY TAKEAWAY

PD shows "how much" (protons).
T1 shows "how fast" (recovery).
Mastering PD and T1 is essential for MRI success.

★ **RIGHT INFORMATION. RIGHT BALANCE. RIGHT IMAGE.** Understand the sequence. Recognize the artifacts. Support the technologist. Deliver the value.